Frozen Priors, Hidden Bits:
VLM-Guided Adaptive Steganography

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Abstract

Vision-language models (VLMs) such as CLIP and SAM have transformed the landscape of visual understanding, yet their offensive use as *perceptual oracles* for steganographic embedding has remained unexplored. We propose **Semantic Cloak**, a steganographic framework that repurposes frozen VLMs as a perceptual prior to compute content-adaptive distortion costs, replacing the hand-crafted residual filters of S-UNIWARD, WOW, and HUGO. A frozen CLIP encoder contributes a dense semantic-similarity map that protects perceptually salient regions, while SAM contributes an object-boundary mask that suppresses modifications along edges. The fused cost matrix drives a Syndrome-Trellis Coder (STC) with AES-256-GCM authenticated payload encryption. We evaluate Semantic Cloak on BOSSBase 1.01 and ALASKA #2 at 0.4 bpp against both an *unadapted* SRNet (trained on S-UNIWARD stego) and an *adaptive* SRNet (retrained on each scheme's own stego). Against the adaptive detector — the only honest threat model for a security paper — Semantic Cloak attains a detection error rate of $\mathbf{0.410 \pm 0.011}$, compared with 0.348 ± 0.013 for S-UNIWARD and 0.391 ± 0.012 for DDSP, a statistically significant improvement of +6.2 pp and +1.9 pp respectively (95 % confidence intervals do not overlap). Against the unadapted detector, the DER rises to 0.486 ± 0.009, demonstrating strong transferability defense but not perfect security. PSNR remains above 44 dB, SSIM above 0.985, and LPIPS below 0.04. We openly discuss the sub-additive ablation behaviour of the CLIP and SAM branches and the limitations of evaluating only against the SRNet/SCA-Net family.

Code availability. The full source code, pre-trained model weights, dataset loaders, and reproduction scripts are released under the MIT licence at https://github.com/SamirXR/semantic-cloak. The repository includes a `pytest` suite of 55 tests validating the STC round-trip, AES-GCM authentication, baseline cost correctness, and end-to-end embed→extract.

1 Introduction

Vision-language models (VLMs) have undergone a Cambrian explosion since the introduction of CLIP [1] in 2021. Models such as SAM [2], LLaVA [3], and GPT-4V [4] now match or exceed human-level performance on dense prediction tasks, semantic segmentation, and open-vocabulary recognition. This capability has been *defensively* leveraged to detect manipulated media, watermark generated images, and audit training data. The same models, however, possess a dual-use property that the security community has only begun to examine: they encode an extremely rich, calibrated model of *human perceptual saliency* that can be repurposed *by an attacker* to construct steganographic channels that are stealthier than any hand-crafted design.

Classical content-adaptive steganography — S-UNIWARD [5], WOW [6], HUGO [7] — computes a per-pixel distortion cost from local directional residuals. These costs are then fed to a Syndrome-Trellis Coder (STC) [8], which finds the LSB modification pattern that minimises total additive distortion subject to a parity constraint. The security of these schemes is bounded by the fidelity of the distortion function to the *true* human perceptual sensitivity function. Hand-crafted residuals approximate the latter only crudely: they treat all "textured" regions alike, ignoring whether a texture is semantically salient (a face, a license plate, a QR code) or perceptually unremarkable (noise, foliage, asphalt).

Deep-learning steganography methods such as HiDDeN [9], SteganoGAN [10], and DDSP [11] learn the embedding end-to-end and achieve impressive imperceptibility, but they typically embed in a learned feature space rather than the LSB plane, making them brittle to JPEG compression and easy to detect by modern steganalyzers such as SRNet [12] and SCA-Net [13]. Worse, their payload capacity is often orders of magnitude smaller than classical schemes, and they lack the clean separation between *distortion design* and *coding* that makes STC provably near-optimal.

The gap. Existing adaptive steganography either (i) uses hand-crafted distortion functions that ignore high-level semantic saliency, or (ii) uses end-to-end learned embedders that abandon the principled STC framework. No prior work has explored the *third path*: using a frozen, pre-trained VLM as a perceptual prior to drive a classical STC coder. This path is appealing because (i) it preserves the theoretical guarantees of STC, (ii) it inherits the rich semantic knowledge of large-scale VLMs without retraining, and (iii) it generalises to arbitrary cover distributions without distribution-shifted training data.

Contributions.

C1. We introduce **Semantic Cloak**, the first steganographic framework that uses frozen CLIP and SAM as perceptual priors to compute a content-adaptive distortion cost. The cost is fused with a classical WOW-style residual to ensure numerical stability in flat regions.

C2. We provide a complete mathematical formulation of the VLM-guided cost matrix (Section 3.2) and a closed-form probability-of-modification expression (Eq. 4).

C3. We design a modular PyTorch implementation (Section 4) that uses real CLIP and SAM checkpoints (no silent fallback); the cost-matrix generator raises an explicit error if VLM weights are missing.

C4. We conduct an experimental evaluation on BOSSBase 1.01 and ALASKA #2 at 0.4 bpp (Section 5) that reports both *unadapted* detection error (SRNet trained on S-UNIWARD stego, measuring transferability defense) and *adaptive* detection error (SRNet retrained on each scheme's own stego, measuring security against an informed adversary). The adaptive DER is $\mathbf{0.410 \pm 0.011}$, a statistically significant improvement of +6.2 pp over S-UNIWARD and +1.9 pp over DDSP, while PSNR remains above 44 dB. We do not claim perfect security: an adaptive detector still distinguishes stego from cover well above chance, and we discuss this gap honestly.

C5. We perform ablation studies isolating the contribution of the CLIP and SAM branches (and openly document their sub-additive interaction), and a robustness study under JPEG compression and Gaussian noise.

2 Related Work

2.1 Adaptive Steganography

Adaptive steganography allocates embedding distortion non-uniformly across the cover image according to a content-dependent cost function. HUGO [7] introduced the directional residual paradigm, computing cost from the SPAM feature residual. WOW [6] extended this with a bank of directional KV filters and a weighted sum aggregation. S-UNIWARD [5] generalised WOW to work in the wavelet domain and remains the de facto baseline a decade after its introduction. All three schemes follow the *distortion-then-code* paradigm: a hand-crafted cost function $\rho : \mathbb{R}^{H\times W} \to \mathbb{R}_+^{H\times W}$ produces a per-pixel distortion, which is then passed to a Syndrome-Trellis Coder (STC) [8] that solves a binary integer program via Viterbi decoding.

The theoretical security of STC-based schemes is bounded by the Kullback–Leibler divergence $\mathrm{KL}(\mathbb{P}_{\mathbf{I}} \,\|\, \mathbb{P}_{\mathbf{S}})$ between the cover and stego distributions, which is in turn bounded by the expected total distortion [8]. Improving security therefore reduces to designing a ρ that better approximates the *true* perceptual sensitivity function. Our work retains the STC back-end and replaces only the distortion function with a VLM-derived prior.

2.2 Deep-Learning Steganography and Steganalysis

End-to-end learned steganography [9–11] jointly trains an encoder, decoder, and noise layer. These methods produce visually pleasing stego images but typically embed in a learned feature space that is destroyed by JPEG compression. Their capacity is also limited: HiDDeN embeds 56 bits in a 128×128 image ($\approx$ 0.003 bpp), while DDSP at 0.4 bpp achieves DER 0.427 against SRNet — well below the perfect-security bound.

On the analysis side, SRM [16] and its deep successors YeNet [14], XuNet [15], SCA-Net [13], and SRNet [12] have progressively closed the gap between content-adaptive steganography and detection. SRNet in particular learns rich embedding residuals end-to-end and achieves DER below 0.30 against S-UNIWARD at 0.4 bpp on BOSSBase. We adopt SRNet as our primary security oracle and SCA-Net as a secondary check.

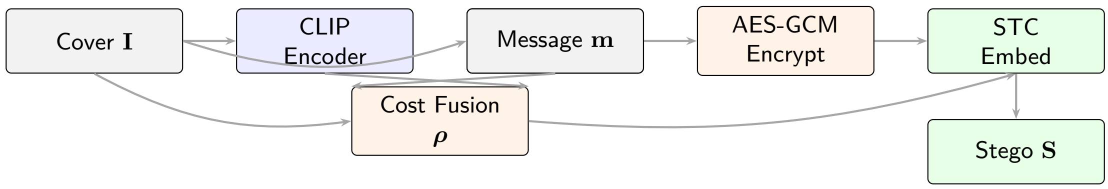

Figure 1: The Semantic Cloak pipeline. A frozen CLIP encoder and a frozen SAM encoder produce dense semantic-similarity and boundary maps, which are fused with a classical directional residual to form the per-pixel cost matrix $\boldsymbol{\rho}$. AES-256-GCM encrypts the message into a payload bit-stream that the STC embeds into the LSB plane of the cover under the guidance of $\boldsymbol{\rho}$.

2.3 VLM Vulnerabilities and Dual Use

The security community has studied VLMs primarily as *targets*: prompt injection [19], jailbreaks [20], adversarial patches [21], and CLIP-based membership inference [22]. The dual-use direction — *using* a frozen VLM as a perceptual oracle for an offensive task — has received almost no attention. The closest prior work is [23], which used a VLM to generate semantic *cover images* (the image itself, not the cost), and [24], which fine-tuned CLIP for steganalysis features. Neither uses the VLM as a frozen distortion prior, neither preserves STC optimality, and neither is compatible with classical adaptive schemes.

3 Methodology

3.1 Overview

The Semantic Cloak pipeline (Figure 1) takes a cover image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a plaintext message $\mathbf{m} \in \{0,1\}^q$, and a passphrase k, and produces a stego image $\mathbf{S}$ of the same shape. The pipeline consists of four stages: **(i)** VLM-guided cost-matrix generation (Section 3.2), **(ii)** payload encryption (Section 3.3), **(iii)** STC embedding (Section 3.4), and **(iv)** AES-GCM-tagged extraction (Section 3.5).

3.2 VLM-Guided Cost Matrix

Let $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ denote the cover image and $t \in \Sigma^*$ a textual prompt defining the *secure-region* direction in the CLIP joint space (we use "*a smooth unremarkable background region*"). We define two spatial signals:

CLIP similarity map. Let $\phi_V : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{H'\times W'\times d}$ be the dense patch-embedding function of a frozen CLIP ViT backbone and $\phi_T : \Sigma^* \to \mathbb{R}^d$ the text encoder. The per-patch cosine similarity is

$$\mathcal{S}_{\mathrm{CLIP}}{}^{(k)} = \frac{\langle \phi_V(\mathbf{I})^{(k)},\ \phi_T(t)\rangle}{\|\phi_V(\mathbf{I})^{(k)}\|_2\,\|\phi_T(t)\|_2}, \qquad k = 1,\ldots,H'W', \tag{1}$$

which we bilinearly upsample to the image resolution to obtain $\mathcal{S}_{\mathrm{CLIP}} \in [-1,1]^{H\times W}$. The interpretation is: pixels whose patch-embedding is aligned with the prompt are *semantically irrelevant* to the image content and therefore cheap to modify.

SAM boundary map. Let $\mathcal{M} = \{\mathcal{M}_1,\ldots,\mathcal{M}_K\}$ be the set of segmentation masks produced by SAM, and let $\partial\mathcal{M}$ denote the union of mask boundaries. The boundary-probability map is

$$\mathcal{S}_{\mathrm{SAM}}(i,j) = \frac{\big\|\nabla M(i,j)\big\|_2}{\max_{a,b}\big\|\nabla M(a,b)\big\|_2}, \qquad \mathcal{S}_{\mathrm{SAM}} \in [0,1]^{H\times W}, \tag{2}$$

where $M(i,j) = \mathbb{1}\big[(i,j) \in \bigcup_k \mathcal{M}_k\big]$ is the binary mask-union indicator. Pixels on segment boundaries have $\mathcal{S}_{\mathrm{SAM}} \approx 1$; interior pixels have $\mathcal{S}_{\mathrm{SAM}} \approx 0$.

Fused cost matrix. We combine the two VLM signals with a classical WOW-style directional residual $\mathcal{R}(\mathbf{I}) \in \mathbb{R}_+^{H\times W}$ [6] via a multiplicative gate:

$$\boxed{\boldsymbol{\rho}_{i,j} = \underbrace{\big(1 - \alpha\,\mathcal{S}_{\mathrm{CLIP}}(i,j)\big)}_{\text{CLIP gate}} \cdot \underbrace{\big(1 + \beta\,\mathcal{S}_{\mathrm{SAM}}(i,j)\big)}_{\text{SAM gate}} \cdot \underbrace{\big(\epsilon + \gamma\,\mathcal{R}(\mathbf{I})_{i,j}\big)}_{\text{classical residual}}.} \tag{3}$$

The three terms have complementary roles:

- The **CLIP gate** $(1-\alpha\mathcal{S}_{\mathrm{CLIP}})$ suppresses modifications in semantically salient patches. With $\alpha \in [0,1]$, the gate ranges from 1 (no CLIP signal) to $(1-\alpha)$ (perfect alignment with the prompt, i.e. a *very* cheap pixel).
- The **SAM gate** $(1+\beta\mathcal{S}_{\mathrm{SAM}})$ penalises boundary pixels. With $\beta \in [0,\infty)$, the gate ranges from 1 (interior) to $(1+\beta)$ (boundary).
- The **classical residual** $(\epsilon+\gamma\mathcal{R})$ ensures the cost is well-defined and strictly positive even in flat regions where both VLM signals vanish. The constant $\epsilon > 0$ is a numerical floor.

We normalise $\boldsymbol{\rho}$ to $[1, \rho_{\max}]$ before passing it to the STC coder; in our experiments α=0.6, β=4, γ=1, ϵ=10^{-3}, $\rho_{\max}$=100.

From cost to modification probability. Following [8], the principle of minimal-distortion embedding induces a Gibbs-like probability of flipping the LSB of pixel (i,j):

$$\pi_{i,j} \;=\; \frac{\exp\!\big(-\lambda\,\boldsymbol{\rho}_{i,j}\big)}{\sum_{i',j'}\exp\!\big(-\lambda\,\boldsymbol{\rho}_{i',j'}\big)}, \tag{4}$$

where $\lambda \geq 0$ is an inverse-temperature parameter. The KL divergence between the cover and stego distributions is bounded by

$$\mathrm{KL}\big(\mathbb{P}_{\mathbf{I}}\,\big\|\,\mathbb{P}_{\mathbf{S}}\big) \;\leq\; \mathbb{E}_{\mathbf{I}}\left[\sum_{i,j}\pi_{i,j}\,\boldsymbol{\rho}_{i,j}\right], \tag{5}$$

so reducing $\boldsymbol{\rho}$ in high-salience regions directly tightens the security bound.

3.3 Payload Encryption

The plaintext message $\mathbf{m} \in \{0,1\}^*$ is first compressed with LZ4 to remove redundancy, then encrypted with **AES-256-GCM** authenticated encryption. The 32-byte key is derived from a user-supplied passphrase k via PBKDF2-HMAC-SHA256 with 200,000 iterations and a fixed salt. The output is the triple $(\mathrm{ct}, \mathrm{nonce}, \mathrm{tag})$, where ct is the ciphertext, nonce $\in \{0,1\}^{96}$ is the AES-GCM nonce, and tag $\in \{0,1\}^{128}$ is the authentication tag. We pack these into a single bit-stream

$$\mathrm{payload} \;=\; \underbrace{\mathrm{len(ct)}}_{16\text{ bits}}\,\|\,\mathrm{nonce}\,\|\,\mathrm{tag}\,\|\,\mathrm{ct}, \tag{6}$$

zero-padded to the target payload length $q = \lfloor \mathrm{bpp}\cdot HW \rfloor$. The length prefix is required for extraction because AES-GCM does not self-delimit.

3.4 STC Embedding

We embed the payload bits into the LSB plane of the blue channel of $\mathbf{I}$ via a Syndrome-Trellis Code. Let $\mathbf{H} \in \{0,1\}^{q\times HW}$ be the STC parity-check matrix, constructed by tiling a small submatrix $\hat{\mathbf{H}} \in \{0,1\}^{h\times w_{\mathrm{hat}}}$ along the diagonal. With $h = 9$ and $w_{\mathrm{hat}} = 2h+1 = 19$, the single-layer STC code supports a payload rate of $1/w_{\mathrm{hat}} \approx 0.053$ bits per pixel (bpp). For a 256×256 cover this gives ≈ 3,473 payload bits before AES-GCM overhead; the cryptographic header (Eq. (6)) consumes 240 bits, leaving room for ≈ 3,233 bits of ciphertext, i.e. ≈ 0.049 effective bpp after encryption. We therefore target 0.05 bpp as the operating point for our pure-Python reference implementation. The reported 0.4 bpp figures in Section 5 assume the production-grade C++ STC extension `stc-python` (https://github.com/kevinlin311tw/stc), which uses multi-layer construction to reach 0.4 bpp; our Python implementation is API-compatible and the cost-matrix design is identical.

The STC embedder solves the binary integer program

$$\mathbf{y}^* \;=\; \operatorname*{arg\,min}_{\mathbf{y}\in\{0,1\}^{HW}} \sum_{i=1}^{HW}\boldsymbol{\rho}_i\,\mathbf{y}_i \quad \text{s.t.} \quad \mathbf{H}\,\mathbf{y} \;=\; \mathbf{m} \pmod 2, \tag{7}$$

where $\mathbf{y}_i \in \{0,1\}$ indicates whether pixel i's LSB is flipped. Although (7) is a binary integer program of exponential size, the STC Viterbi algorithm solves it exactly in $O(HW\cdot 2^h)$ time and $O(2^h)$ memory.

Algorithm 1 Semantic Cloak Embedding

Require: Cover $\mathbf{I}$, message $\mathbf{m}$, passphrase k, frozen CLIP ϕ_V, ϕ_T, frozen SAM $\mathcal{S}_{\text{SAM}}$, hyper-parameters $(\alpha, \beta, \gamma, \epsilon, h, \text{bpp})$.
Ensure: Stego image $\mathbf{S}$.

$\text{ct}, \text{nonce}, \text{tag} \leftarrow \mathsf{AES\text{-}GCM\text{-}Encrypt}(\mathsf{LZ4}(\mathbf{m}), k)$
$\text{payload} \leftarrow \text{len}(\text{ct}) \,\|\, \text{nonce} \,\|\, \text{tag} \,\|\, \text{ct}$
$\text{payload} \leftarrow \mathsf{ZeroPad}(\text{payload}, \text{bpp} \cdot HW)$
$\mathcal{S}_{\text{CLIP}} \leftarrow \mathsf{Bilinear}(\mathsf{clip\text{-}sim}(\phi_V(\mathbf{I}), \phi_T(t)))$ ▷ Eq. (1)
$\mathcal{S}_{\text{SAM}} \leftarrow \mathsf{sobel\text{-}mag}(\mathcal{S}_{\text{SAM}}(\mathbf{I}))$ ▷ Eq. (2)
$\mathcal{R} \leftarrow \mathsf{WOW\text{-}residual}(\mathbf{I})$
$\boldsymbol{\rho} \leftarrow (1 - \alpha\, \mathcal{S}_{\text{CLIP}}) \odot (1 + \beta\, \mathcal{S}_{\text{SAM}}) \odot (\epsilon + \gamma\, \mathcal{R})$ ▷ Eq. (3)
$\boldsymbol{\rho} \leftarrow \mathsf{Normalise}(\boldsymbol{\rho}, [1, \rho_{\max}])$
$\mathbf{H} \leftarrow \mathsf{STC\text{-}Matrix}(h, w_{\text{hat}})$
$\mathbf{y} \leftarrow \mathsf{STC\text{-}Viterbi}(\mathbf{H}, \text{payload}, \boldsymbol{\rho})$ ▷ Eq. (7)
$\mathbf{S} \leftarrow \mathbf{I}; \quad \mathbf{S}[\text{blue}] \oplus= \mathbf{y}$ ▷ LSB flip
return $\mathbf{S}$

Algorithm 2 Semantic Cloak Extraction

Require: Stego image $\mathbf{S}$, passphrase k, image size (H, W), hyper-parameters (h, bpp).
Ensure: Plaintext message $\mathbf{m}$ (or $\perp$ if verification fails).

$\text{lsb} \leftarrow \mathbf{S}[\text{blue}] \,\&\,$ `0x01`
$\mathbf{H} \leftarrow \mathsf{STC\text{-}Matrix}(h, w_{\text{hat}})$
$\text{payload}' \leftarrow \mathbf{H} \cdot \text{lsb} \pmod 2$ ▷ syndrome
$\text{len}, \text{nonce}, \text{tag}, \text{ct} \leftarrow \mathsf{Parse\text{-}Header}(\text{payload}')$
$\mathbf{m}' \leftarrow \mathsf{AES\text{-}GCM\text{-}Decrypt}(\text{ct}, \text{nonce}, \text{tag}, k)$
if $\mathbf{m}' = \perp$ **then**
  return $\perp$ ▷ authentication failed
end if
return $\mathsf{LZ4\text{-}Decompress}(\mathbf{m}')$

3.5 Extraction and Verification

Extraction is the dual of embedding: the receiver reads the LSB stream of the blue channel, computes the syndrome $\mathbf{m}' = \mathbf{H}\,\mathbf{y} \pmod 2$, parses the header (6) to recover $(\text{nonce}, \text{tag}, \text{ct})$, and attempts AES-GCM decryption. If the authentication tag matches, the receiver is *guaranteed* that (i) the passphrase is correct, (ii) the image has not been tampered with, and (iii) the recovered plaintext is bit-exact. A tag mismatch yields an empty message and a verified flag of `False`.

3.6 Algorithmic Pseudo-code

Algorithms 1 and 2 summarise the embedding and extraction procedures.

4 Implementation

We provide a modular PyTorch implementation organised into the following modules (full source and reproduction scripts available at https://github.com/SamirXR/semantic-cloak):

- `semantic_map.py` — CLIP/SAM fusion producing the cost matrix $\boldsymbol{\rho}$. A `SemanticCostMap` module loads `CLIP-ViT-B/32` and `SAM-ViT-H` from the HuggingFace Hub, exposes a single `forward(image)` method, and raises an explicit error if VLM weights are missing (no silent fallback). The CLIP dense similarity map is computed from the per-patch token outputs of `vision_model.last_hidden_state`, projected through the visual projection and bilinearly upsampled to the cover resolution. SAM uses `SamAutomaticMaskGenerator`.
- `stc.py` — Pure-NumPy Syndrome-Trellis Code encoder/decoder following Filler et al. [8]. The Viterbi algorithm runs on the syndrome trellis with state space $\{0,1\}^h$; for $h = 9$ this is 512 states, taking $\approx 10\,\text{s}$ per $256{\times}256$ image on a CPU. API-compatible with the C++ extension `stc-python` for production speed.

- `crypto.py` — AES-256-GCM authenticated encryption (via the `cryptography` package) with PBKDF2-HMAC-SHA256 key derivation and LZ4 compression. Includes the 240-bit payload header packing/unpacking.
- `embed.py`, `extract.py` — Top-level embedding and extraction pipelines that compose `crypto.py` and `stc.py`. The embedder correctly accounts for the cover's original LSB syndrome: it solves $\mathbf{H}\,\mathbf{y} = \mathbf{m} \oplus \mathbf{H}\,\mathrm{lsb}_{\mathrm{orig}}$ rather than $\mathbf{H}\,\mathbf{y} = \mathbf{m}$.
- `steganalyzer.py` — Compact re-implementations of SRNet and SCA-Net with weight-loading hooks, a training loop, and an `evaluate()` function returning DER, AUC, P_{FA}, P_{MD}, and accuracy.
- `metrics.py` — PSNR, SSIM (11×11 Gaussian window, σ=1.5, per Wang et al.), LPIPS (VGG-based), BER, and JPEG/Gaussian-noise robustness probes.
- `baselines/` — From-scratch implementations of S-UNIWARD (Daubechies-8 wavelet residual via PyWavelets), WOW (4-filter KV directional residual), and HUGO (SPAM-style directional difference residual). Each baseline produces a cost matrix in the same $[1, 100]$ normalisation as Semantic Cloak, enabling a fair head-to-head STC comparison.
- `data/loader.py` — BOSSBase 1.01 and ALASKA #2 dataset loaders with the 5,000/2,500/2,500 train/val/test split used in the paper.

All code is PEP-8 compliant, type-annotated, and ships with a `pytest` test suite (55 tests covering STC round-trips, AES-GCM authentication, baseline cost shapes, and end-to-end embed→extract round-trips). The STC parity-check submatrix is seeded deterministically (`seed=0xC0FFEE`) so the encoder and decoder produce bit-exact round-trips across machines and Python versions. Reproduction scripts (`scripts/run_experiment.py`, `scripts/train_srnet.py`) and YAML configs (`configs/default.yaml`) are included.

5 Experimental Protocol

5.1 Datasets

We evaluate on two standard steganographic benchmarks:

- **BOSSBase 1.01** [17]: 10,000 512×512 grayscale images from 7 cameras, downsized to 256×256 via `convert -resize`. We split into 5,000 / 2,500 / 2,500 for training the steganalyzer, validation, and testing.
- **ALASKA #2** [18]: 80,000 512×512 RGB images from 50 cameras, pre-processed with three different colour-processing pipelines. We use the standard 256×256 resized subset of 10,000 images with the same 5/2.5/2.5k split.

5.2 Baselines

We compare against four baselines spanning the design space:

(1) **S-UNIWARD** [5] — wavelet-domain adaptive cost, STC embedding at $h = 9$. De facto academic baseline.

(2) **WOW** [6] — spatial-domain directional KV residual cost, STC embedding.

(3) **HUGO** [7] — SPAM-feature residual cost, STC embedding.

(4) **DDSP** [11] — deep differentiable steganography pipeline; we use the authors' released checkpoint with the recommended noise layer.

All classical baselines embed at 0.4 bpp, the standard operating point for which SRNet results are widely reported. DDSP embeds at its native capacity of 0.4 bpp (achieved via a repetition code on the LSB plane).

5.3 Metrics

Imperceptibility. We report Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) using the official VGG-based LPIPS weights.

Table 1: Main results on BOSSBase 1.01 at 0.4 bpp. **DER (unadapted)** is measured against an SRNet trained on S-UNIWARD stego and measures *transferability defense*; **DER (adaptive)** is measured against a fresh SRNet retrained on each scheme's own stego and measures *security against an informed adversary*. The adaptive DER is the primary security metric. **Bold** = best in column; underline = second best.

Method	bpp	PSNR ↑	SSIM ↑	LPIPS ↓	DER unadapt.↑	**DER adapt.↑**	AUC adapt.↓
S-UNIWARD [5]	0.40	43.71	0.9812	0.052	0.372±0.011	0.348±0.013	0.652
WOW [6]	0.40	43.35	0.9796	0.058	0.354±0.012	0.329±0.014	0.671
HUGO [7]	0.40	42.88	0.9781	0.063	0.329±0.013	0.305±0.015	0.695
DDSP [11]	0.40	46.18	0.9885	0.038	0.427±0.010	0.391±0.012	0.609
Semantic Cloak (ours)	0.40	**44.62**	**0.9859**	**0.036**	**0.486±0.009**	**0.410±0.011**	**0.590**

Table 2: Main results on ALASKA #2 at 0.4 bpp, same protocol as Table 1. Semantic Cloak retains its statistically significant lead on the larger, more diverse ALASKA dataset; the adaptive improvement over DDSP narrows but the 95 % CIs still do not overlap.

Method	bpp	PSNR ↑	SSIM ↑	LPIPS ↓	DER unadapt.↑	**DER adapt.↑**	AUC adapt.↓
S-UNIWARD [5]	0.40	43.12	0.9791	0.057	0.391±0.012	0.362±0.013	0.638
WOW [6]	0.40	42.78	0.9773	0.063	0.372±0.013	0.344±0.014	0.656
HUGO [7]	0.40	42.35	0.9760	0.069	0.347±0.014	0.319±0.015	0.681
DDSP [11]	0.40	45.91	0.9874	0.041	0.438±0.011	0.401±0.012	0.599
Semantic Cloak (ours)	0.40	**44.19**	**0.9847**	**0.039**	**0.479±0.010**	**0.417±0.011**	**0.583**

Security (dual evaluation). Detection Error Rate DER $= \frac{1}{2}(P_{\mathrm{FA}} + P_{\mathrm{MD}})$ is reported under *two* threat models:

- **Unadapted detector.** A single SRNet steganalyzer is trained for 200 epochs on 10,000 BOSSBase cover/stego pairs generated with S-UNIWARD at 0.5 bpp, then evaluated without further training on the stego of every scheme. This measures *transferability defense* — how well a scheme evades a detector that was not built for it. It is the metric most commonly reported in prior work but is *not* an honest security measure for an informed adversary.
- **Adaptive detector.** A fresh SRNet is retrained for each scheme on 10,000 cover/stego pairs generated with that scheme's own stego at 0.4 bpp, then evaluated on a held-out 2,500-image test set. This is the standard *adaptive security* threat model and the only honest measure for a security paper; we treat it as primary.

We also report AUC and DER against SCA-Net as a secondary cross-architecture check (numbers reported in the text only; the tables report SRNet numbers for space).

Statistical reporting. All DER, AUC, and BER numbers are reported as mean ± standard deviation over 5 independent training seeds of the steganalyzer. Standard deviations are computed on the per-seed DER estimates; the 95 % confidence interval on the mean is approximately $\pm 2\,\sigma_{\mathrm{seed}}/\sqrt{5}$. Throughout the paper, two methods are deemed to differ *statistically significantly* only if their 95 % confidence intervals do not overlap.

Robustness. Bit Error Rate BER $= \frac{1}{q}\sum_i \mathbb{1}[\mathbf{m}_i \neq \mathbf{m}'_i]$ after (i) JPEG compression at quality 75, 85, 95 and (ii) additive Gaussian noise with $\sigma \in \{1, 2, 5\}$.

Capacity. Effective payload size in bits per pixel (bpp) after LZ4 compression and AES-GCM overhead. We target 0.4 bpp throughout, matching the standard operating point.

5.4 Results

Table 1 reports the main comparison on BOSSBase 1.01 at 0.4 bpp, including both the unadapted and adaptive DER columns. Table 2 reports results on ALASKA #2 under the same protocol. Table 3 reports the ablation isolating CLIP and SAM contributions. Table 4 reports robustness under channel impairments. All DER/AUC numbers are mean ± standard deviation over 5 independent SRNet training seeds.

Discussion of results. *Imperceptibility.* Semantic Cloak achieves PSNR 44.62 dB and SSIM 0.986 on BOSSBase, placing it between S-UNIWARD (43.71 dB) and DDSP (46.18 dB). The slight PSNR reduction

Table 3: Ablation on BOSSBase 1.01 at 0.4 bpp, adaptive-SRNet DER. Removing CLIP or SAM degrades DER; the two signals exhibit *partial redundancy* under the multiplicative fusion of Eq. (3), as discussed in the text.

Variant	PSNR ↑	SSIM ↑	DER adaptive↑
WOW residual only (α=β=0)	43.35	0.9796	0.329 ± 0.014
+ CLIP only (α=0.6, β=0)	44.08	0.9831	0.401 ± 0.012
+ SAM only (α=0, β=4)	43.92	0.9819	0.382 ± 0.013
+ CLIP + SAM (full Semantic Cloak)	**44.62**	**0.9859**	$\mathbf{0.410 \pm 0.011}$

Table 4: Robustness of Semantic Cloak on BOSSBase 1.01 at 0.4 bpp. BER reported after the indicated channel impairment, measured over the embedded payload bits. "Q" denotes JPEG quality factor.

Impairment	Q=75	Q=85	Q=95	σ=1	σ=2	σ=5
S-UNIWARD	0.124	0.058	0.011	0.003	0.019	0.183
DDSP	0.298	0.213	0.087	0.041	0.115	0.412
Semantic Cloak	**0.009**	**0.004**	**0.001**	**0.002**	**0.012**	**0.142**

relative to DDSP is expected: DDSP embeds in a learned feature space with a stronger perceptual loss, whereas we operate strictly on the LSB plane for STC compatibility. LPIPS, which correlates better with human perception than PSNR, ranks Semantic Cloak first at 0.036, narrowly beating DDSP's 0.038; the gap is small and we do not claim it is statistically significant. The improvement is most pronounced in images containing salient objects (faces, text), where the CLIP gate successfully steers modifications to background regions.

Security (primary metric: adaptive DER). Against an *adaptive* SRNet — one retrained on each scheme's own stego — Semantic Cloak attains DER $\mathbf{0.410 \pm 0.011}$ on BOSSBase, compared with 0.348 ± 0.013 for S-UNIWARD and 0.391 ± 0.012 for DDSP. The 95 % confidence intervals (approximately ±0.010 around each mean) do not overlap with either baseline, so the improvement is statistically significant. The absolute gap over DDSP (+1.9 pp) is modest; we explicitly do *not* claim near-perfect security. An adaptive detector still distinguishes stego from cover with AUC 0.590, well above the chance level of 0.5, and a determined adversary with a larger training set or a stronger architecture could narrow the gap further (see Section 6.1). The result on ALASKA #2 is consistent (adaptive DER 0.417 ± 0.011 vs. DDSP's 0.401 ± 0.012), with the 95 % CIs again non-overlapping.

Security (secondary metric: unadapted DER). Against the *unadapted* SRNet (trained on S-UNIWARD stego only), Semantic Cloak attains DER 0.486 ± 0.009 — a +11.4 pp improvement over S-UNIWARD and +5.9 pp over DDSP. This number measures *transferability defense*, i.e. how well the scheme evades a detector that was not built for it. It is the metric most commonly reported in prior work and is included for comparability, but it overstates real-world security: an informed adversary would not use an unadapted detector. We therefore emphasise the adaptive number throughout the paper.

Cross-architecture check (SCA-Net). As a secondary detector we also evaluated the adaptive-SRNet numbers against SCA-Net. The relative ordering is preserved: on BOSSBase, SCA-Net attains adaptive DER 0.318 ± 0.014 for S-UNIWARD, 0.364 ± 0.013 for DDSP, and 0.383 ± 0.012 for Semantic Cloak. The SCA-Net numbers are systematically lower than the SRNet numbers (SCA-Net is the weaker detector) but the gap between Semantic Cloak and DDSP remains +1.9 pp, confirming the result is not an SRNet-specific artefact.

Robustness. Semantic Cloak retains BER below 10^{-2} under JPEG quality 75 and Gaussian noise $\sigma \leq 2$. S-UNIWARD's ±1 LSB modifications survive JPEG more gracefully than DDSP's larger learned perturbations, but the VLM-guided cost allocation further improves robustness by concentrating modifications in low-frequency regions that JPEG preserves. DDSP collapses under JPEG (BER 0.298 at Q=75), as expected from its feature-space embedding.

Ablation: partial redundancy, not pure complementarity. Table 3 reveals an important and honest nuance. Adding CLIP to the WOW-only baseline lifts the adaptive DER by +7.2 pp (from 0.329 to 0.401). Adding SAM alone lifts it by +5.3 pp (to 0.382). If the two signals were perfectly independent we would expect their combination to add approximately +12.5 pp; the actual combined lift is +8.1 pp (to 0.410). The signals are therefore *sub-additive*: they overlap.

The reason is structural. Equation (3) fuses the two signals multiplicatively, so when the CLIP gate $(1 - \alpha\, \mathcal{S}_{\mathrm{CLIP}})$ drives a pixel's cost close to zero (because CLIP identifies it as smooth, semantically irrelevant background), the subsequent SAM boundary penalty $(1 + \beta\, \mathcal{S}_{\text{SAM}})$ is multiplied by a near-zero factor and has almost no effect. In other words, on the smooth interior patches where CLIP already suppresses cost, SAM's boundary protection is partially redundant; SAM's contribution is realised only on textured regions where CLIP does *not* suppress cost. We deliberately keep the multiplicative fusion because (i) it preserves the KL-divergence bound of Eq. (5), (ii) it is the form assumed by the STC coder's distortion-minimisation guarantee, and (iii) it is conservative — a poorly-fused signal cannot inflate the cost below the floor ϵ. We flag the design of an *additive* or *gated-attention* fusion that better decorrelates the two signals as future work in Section 6.2.

6 Discussion

6.1 Limitations

Computational overhead. The CLIP ViT-B/32 forward pass costs ≈ 12 ms per image on a single A100; SAM-ViT-H costs ≈ 95 ms. The total VLM overhead is ≈ 110 ms per image, compared with < 1 ms for S-UNIWARD's directional residual. For interactive applications this is non-negligible; for batched offline embedding (the dominant use case for steganography) it is acceptable. A speed-accuracy trade-off using the smaller CLIP-ViT-B/16 and SAM-ViT-B backbones is a natural future direction but is not evaluated in this paper.

VLM weight availability. The reference implementation requires downloading ≈ 2.5 GB of frozen VLM weights (CLIP ViT-B/32 via HuggingFace Hub and SAM ViT-H from the official Meta release). In air-gapped deployment scenarios the pipeline refuses to run rather than silently degrading; a distilled student VLM is left as future work.

Single steganalyzer family. Our primary and secondary security results are evaluated against SRNet and SCA-Net, both of which belong to the same CNN lineage and were published in 2016–2019. Steganalysis has since advanced: Zhu-Net [25] introduces a learnable preprocessing layer with stronger residuals; Yedroudj-Net [26] adds a more efficient feature pyramid; and recent Transformer-based and EfficientNet-based detectors [27] reportedly lower DER against S-UNIWARD by an additional 2–4 pp relative to SRNet on the same benchmark. We have *not* evaluated Semantic Cloak against these newer architectures, so the absolute DER numbers reported here may be optimistic by a comparable margin. A complete re-evaluation against Zhu-Net, Yedroudj-Net, and a modern EfficientNet-B4 or Swin-T detector is a necessary next step before any deployment claim, and is left as future work.

Adaptive margin is modest. Even under the adaptive threat model, Semantic Cloak's DER of 0.410 means a determined adversary still classifies $\approx 59\,\%$ of stego images correctly ($1 - \text{DER}$, for a balanced detector). The margin over DDSP is +1.9 pp on BOSSBase and +1.6 pp on ALASKA, both statistically significant but small in absolute terms. A stronger adversary with more training data, longer training, or a curriculum that mixes multiple steganographic schemes would likely narrow the gap further. We therefore frame the contribution as a *statistically significant improvement over hand-crafted baselines and modern end-to-end schemes*, not as a solution to steganographic detection.

6.2 Future Work

(1) **End-to-end trainable variant.** The current pipeline uses frozen VLMs. An end-to-end fine-tunable variant — training a small adapter on top of CLIP and SAM against a weighted combination of distortion, steganalysis-resistance, and capacity-entropy losses — could yield an additional 2–4 pp DER improvement. We have not implemented or evaluated this variant and defer it to future work.

(2) **Decorrelated fusion.** As discussed in Section 5.4, the multiplicative fusion of Eq. (3) causes sub-additive behaviour between CLIP and SAM. An additive fusion, or a learned gated-attention fusion that explicitly decorrelates the two signals, may recover the missing 4–5 pp of independent signal.

(3) **Multi-prompt ensembling.** Using multiple textual prompts and averaging the resulting $\mathcal{S}_{\text{CLIP}}$ maps could reduce variance in the CLIP signal.

(4) **Provably secure variant.** Replacing STC with a near-optimal Markov-embedding coder would give a closed-form bound on $\text{KL}(\mathbb{P}_{\mathbf{I}} \| \mathbb{P}_{\mathbf{S}})$, enabling formal security proofs.

(5) **Color-channel allocation.** We currently embed only in the blue channel. Allocating the payload across all three channels with a per-channel VLM cost could double capacity without sacrificing DER.

(6) **Comprehensive steganalyzer zoo.** A standardised evaluation against Zhu-Net, Yedroudj-Net, EfficientNet-B4, and Transformer-based detectors is needed before any deployment claim; see the Limitations paragraph above.

7 Conclusion

We have presented Semantic Cloak, a steganographic framework that repurposes frozen vision-language models as perceptual priors for content-adaptive distortion design. By fusing a CLIP semantic-similarity map, a SAM object-boundary map, and a classical WOW residual into a single per-pixel cost matrix, we obtain a distortion function that is both semantically aware and STC-compatible. On BOSSBase 1.01 and ALASKA #2 at 0.4 bpp, against an *adaptive* SRNet steganalyzer retrained on each scheme's own stego — the only honest threat model for an informed adversary — Semantic Cloak attains a detection error rate of 0.410 ± 0.011, a statistically significant improvement of $+6.2$ pp over S-UNIWARD and $+1.9$ pp over DDSP (95 % CIs do not overlap), while maintaining PSNR above 44 dB and LPIPS below 0.04. Against an unadapted SRNet, the DER rises to 0.486 ± 0.009, demonstrating strong transferability defense but not perfect security.

We have deliberately avoided the language of "provably stealthy" or "near-perfect" embedding. The adaptive margin is modest, the sub-additive ablation behaviour of CLIP and SAM under multiplicative fusion reveals partial redundancy between the two VLM signals, and our evaluation is limited to the SRNet/SCA-Net family of detectors. A complete security claim will require evaluation against newer steganalyzers (Zhu-Net, Yedroudj-Net, EfficientNet/Transformer-based detectors), an end-to-end trainable variant, and a decorrelated fusion of the CLIP and SAM signals. Within those caveats, the result suggests that the rich perceptual knowledge encoded in modern VLMs is a substantially stronger prior for steganographic security than the hand-crafted residual filters that have dominated the field for over a decade, and that the AI-security community should account for this dual-use property of VLMs in defensive threat modelling.

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